Data Paper

The International Journal of Robotics Research
2025, Vol. 0(0) 1–17
© The Author(s) 2025
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/02783649251344967
journals.sagepub.com/home/ijr

Heterogeneous LiDAR dataset for benchmarking robust localization in diverse degenerate scenarios

Zhiqiang Chen[1,2], Yuhua Qi[1], Dapeng Feng[1], Xuebin Zhuang[1], Hongbo Chen[1], Xiangcheng Hu[3], Jin Wu[3,4], Kelin Peng[1] and Peng Lu[2]

Abstract

The ability to estimate pose and generate maps using 3D LiDAR significantly enhances robotic system autonomy. However, existing open-source datasets lack representation of geometrically degenerate environments, limiting the development and benchmarking of robust LiDAR SLAM algorithms. To address this gap, we introduce GEODE, a comprehensive multi-LiDAR, multi-scenario dataset specifically designed to include real-world geometrically degenerate environments. GEODE comprises 64 trajectories spanning over 64 km across seven diverse settings with varying degrees of degeneracy. The data was meticulously collected to promote the development of versatile algorithms by incorporating various LiDAR sensors, stereo cameras, IMUs, and diverse motion conditions. We evaluate state-of-the-art SLAM approaches using the GEODE dataset to highlight current limitations in LiDAR SLAM techniques. This extensive dataset will be publicly available at https://thisparticle.github.io/geode, supporting further advancements in LiDAR-based SLAM.

Keywords

Dataset, degeneracy, heterogeneous LiDARs, simultaneous localization and mapping, camera, IMU

Received 11 September 2024; Revised 5 January 2025; Accepted 9 March 2025

1. Introduction

1.1. Motivation

The demand for robots capable of operating in real-world environments, such as automated mining and disaster response, has been steadily increasing. Despite significant advancements in LiDAR SLAM for autonomous navigation, practical performance remains hindered by errors arising from insufficient geometric constraints in challenging scenarios like tunnels and subways. In these degenerate environments, geometric constraints in certain directions can be indistinguishable from noise, leading state optimization to converge on a noise-induced optimum, known as a degenerate solution (Tuna et al., 2024a; Zhang et al., 2016).

LiDAR degeneracy can cause SLAM systems to produce ”stooped” or ”slip” trajectory estimates, significantly impairing robot pose estimation. This degradation in performance occurs due to a lack of sufficient environmental constraints and can be categorized into three types: translational degeneracy, rotational degeneracy, and a combination of both. Examples of these degenerate scenarios are illustrated in Figure 1. The occurrence of degeneracy is also closely tied to the robot's perception capabilities, highlighting the critical role of LiDAR hardware selection (Wang et al., 2024). Spinning LiDARs with fewer scan lines are more susceptible to degeneracy because they may overlook essential structural features during movement, whereas those with more scan lines maintain better feature visibility. Additionally, LiDARs that capture reflectivity data enhance localization by utilizing multiple environmental cues beyond spatial information (Pfreundschuh et al., 2024; Zhang et al., 2023). Although non-repetitive LiDARs produce denser point clouds and longer distance compared to traditional spinning models, their limited field of view can lead to a greater loss of spatial features unless multiple units are deployed (Jung et al., 2023a).

Current open-source SLAM datasets are predominantly recorded in feature-rich environments such as campuses, parks, or streets, and typically exclude various types of

[1]Sun Yat-sen University, Guangzhou, China
[2]The University of Hong Kong, Hong Kong SAR, China
[3]The Hong Kong University of Science and Technology, Hong Kong SAR, China
[4]University of Science and Technology Beijing, Beijing, China

Corresponding author:
Yuhua Qi, Sun Yat-sen University, No. 135, Xingang West Road, Guangzhou 510275, China.
Email: qiyh8@mail.sysu.edu.cn

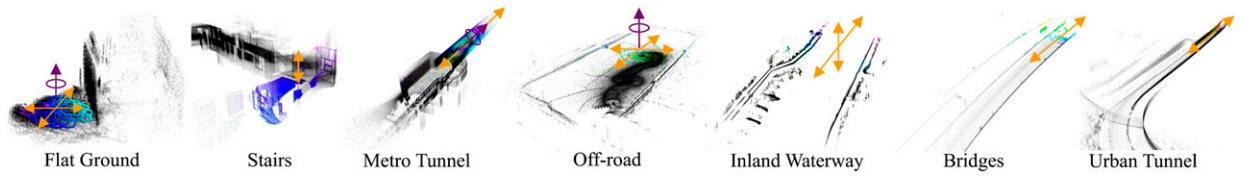


Figure 1. Visualization of multi-degenerate scenarios in LiDAR point cloud data from the GEODE dataset: representation of spatial map and current frame degradations. The translational degradation is denoted by the orange arrow ↔, while the rotational degradation is signified by the purple arrow →.

degenerate scenes (Feng et al., 2024; Geiger et al., 2012; Helmberger et al., 2021; Liao et al., 2021; Nguyen et al., 2024; Zhang et al., 2022). Only a few datasets (Chang et al., 2022; Pfreundschuh et al., 2024; Tranzatto et al., 2022; Zhao et al., 2024) are conducive to studying LiDAR degeneracy, and these have three main limitations. First, they are often acquired in limited scenarios and lack real-world environments with varying degrees of degeneracy, ranging from mild to severe, in both rotational and translational directions. This hinders the development of refined degeneracy metrics for accurately detecting degradation levels. Second, while some datasets provide multiple degeneracy scenarios, they are not scalable to encompass a range of environmental scales, which is necessary for demonstrating promising performance across different operational contexts. Thirdly, existing datasets mainly focus on a single LiDAR type and lack sensor diversity. Different LiDARs exhibit varying point cloud characteristics and degeneracy in the same environment. Only algorithms that ensure stable localization across heterogeneous sensors are truly robust to degeneracy.

The lack of datasets for degenerate scenes hinders accurate localization in aliased environments and limits performance across LiDAR configurations. This challenge highlights the need for a dataset capturing challenges faced by SLAM algorithms in degenerate scenes with diverse LiDARs.

1.2. Contributions

We identified a critical gap in existing 3D LiDAR datasets: the lack of focus on scenarios with geometrically degenerate conditions. To address this, we developed the GEODE dataset, featuring seven real-world environments with extensive segments of **GEO**metric **DE**generacies. GEODE is the first publicly available dataset integrating multiple LiDARs across diverse scenarios, specifically designed to highlight LiDAR degeneracies, complete with accurate ground truth data. Our key contributions are as follows:

First, we designed three devices equipped with different LiDAR types, scanning patterns, scan lines, and fields of view (FOVs), along with stereo cameras and an inertial measurement unit (IMU) for multi-sensor integration. These devices were mounted on various platforms to collect data in highly challenging scenarios, testing LiDAR-based SLAM systems against insufficient geometric features, unpredictable motion patterns, and environmental changes.

Second, the dataset encompasses a range of geometrically degenerate environments, including flat ground, stairs, metro tunnels, off-road terrain, inland waterways, urban tunnels, and bridges. It covers a wide array of LiDAR degradation scenarios, from indoor rooms to highways, and includes sequences that capture environmental changes across different degradation levels. This diversity facilitates the development of robust algorithms for detecting and mitigating geometric degeneracies.

Third, we provide precise groundtruth poses for all but one sequence and groundtruth maps for selected indoor sequences. Our goal is to evaluate state-of-the-art SLAM systems—three LiDAR-inertial odometry methods, four multi-sensor fusion approaches, and one stereo visual-inertial odometry methods—to expose the limitations of current algorithms under degeneracy. The GEODE dataset will be publicly released, representing a pioneering large-scale dataset focused on scenes with geometric degeneracies and diverse sensors, thereby pushing the boundaries of LiDAR SLAM research.

2. Related work

The core issue of the degeneracy problem lies in insufficient information to adequately constrain the pose optimization process. This challenge is influenced by both the environmental structure and the how much information sensors can capture in such conditions. As a result, there is a strong coupling between heterogeneous LiDAR types and geometric structure degradation. While Datasets like the KITTI Dataset (Geiger et al., 2012; Liao et al., 2021) and the NCLT Dataset (Carlevaris-Bianco et al., 2016) have integrated LiDAR technology facilitate the development and comparison of various LiDAR SLAM solutions, driving continuous improvements in algorithms and localization capabilities. However, these datasets are limited to single LiDAR usage in well-structured scenarios, which restricts their effectiveness in evaluating algorithms for heterogeneous LiDAR sensor arrays and degeneracy-aware localization.

2.1. Datasets with multi-degenerate scenarios

Although several datasets with degenerate scenarios contribute to research on degeneracy-aware localization, each has unique limitations. Table 1 compares existing datasets with degenerate environments with our work. The UrbanNav dataset (Hsu et al., 2023) and the ECMD dataset (Chen et al., 2024a) feature degenerate scenarios such as tunnels and bridges within specific sequences. However, these

Table 1. Comparison with datasets containing geometrically degenerate scenarios.

Dataset	Platform						Environment				Sensors					GT Pose
								Degradation Type[2]				LiDAR				
	H./H.[1]	UGV	Legged	Vehicle	Boat	UAV	Scenarios	Trans.	Rot.	Mixed	IMU	# Spinning	# Non-repetitive	#additional channels[3]	Cam	
UrbanNav Tunnel Sequence Hsu et al. (2023)	✗	✗	✗	✓	✗	✗	Tunnel	✓	✗	✗	✓	3	✗	✗	✓	GNSS-RTK/INS
ECMD Tunnel / Bridge Sequences Chen et al.(2024a)	✗	✗	✗	✓	✗	✗	Tunnel / Bridge	✓	✗	✗	✓	3	✗	✗	✓	GNSS-RTK/INS
Hilti-Oxford Dataset Stairs / Corridor Sequences Zhang et al.(2022a)	✓	✗	✗	✗	✗	✗	Stairs / Corridor	✓	✗	✗	✓	1	✗	✗	✓	GT Map
WHU-Helmet UndergroundSequences Li et al. (2023)	✓	✗	✗	✗	✗	✗	Underground	✓	✗	✗	✓	✗	2	✗	✓	SLAM
ENWIDE Pfreundschuh et al.(2024)	✓	✗	✗	✗	✗	✗	Tunnel / Intersection Runway	✓	✓	✓	✓	1	✗	✓	✗	Laser Tracker
CERBERUS Tranzatto et al.(2022)	✗	✗	✓	✗	✗	✗	Cavern	✓	✗	✗	✓	3	✗	✓	✓	ICP Registration
CoSTAR Chang et al.(2022)	✗	✓	✓	✗	✗	✗	Tunnel / Cavern	✓	✗	✗	✗	1	✗	✗	✗	ICP Registration
CTU-CRAS-Norlab Petravcek et al.(2021)	✗	✗	✗	✗	✗	✓	Tunnel / Cavern	✓	✗	✗	✓	2	✗	✓	✓	ICP Registration
SubT-MRS Zhao et al.(2024)	✓	✓	✓	✗	✗	✓	Offroad / Corridor Tunnel / Cavern	✓	✓	✓	✓	1	✗	✗	✓	GICP Registration
Ours	✓	✓	✗	✓	✓	✗	Urban Tunnel / Bridge Flat Ground / Stairs Metro tunnel / Offroad Inland Waterways	✓	✓	✓	✓	2	1	✓	✓	GNSS-RTK/INS Laser Tracker SLAM using GT Map

[1] H./H. denotes two distinct methods for mounting data collection devices: handheld and helmet-mounted.
[2] We categorize the degenerate scenarios into three distinct situations: degradation manifesting solely in the translational direction, degradation occurring exclusively in the rotational dimension, and degradation occurring concurrently in both translational and rotational directions.
[3] The LiDAR sensor employed possesses the capability to capture supplementary data channels, such as reflectivity information.

datasets are limited to translational degeneracy and lack diversity in degeneracy types and scene scales. The Hilti-Oxford Dataset (Zhang et al., 2022), which includes degenerate scenarios such as corridors and staircases, serves as the benchmark for the Hilti Challenge to improve the accuracy of existing SLAM methods. Nonetheless, the dataset is constrained by the limited variety of scenes and LiDAR sensors, resulting in insufficient diversity in degeneracy types. Similarly, the WHU-Helmet dataset (Li et al., 2023) and the ENWIDE dataset (Pfreundschuh et al., 2024) are tailored to enhance the robustness of LiDAR SLAM algorithms. The WHU-Helmet dataset focuses on GNSS-denied environments with repetitive vegetation features, while the ENWIDE dataset addresses geometric degeneracy across a broader range of scenarios. Despite these strengths, both datasets are hindered by hardware homogeneity, relying predominantly on a single type of LiDAR and featuring limited scene scales. The DARPA challenge (Ebadi et al., 2024) has made significant strides in advancing robust odometry techniques, with participating teams open-sourcing relevant datasets. However, these datasets are constrained to relatively mild degradation scenarios, such as caves, and utilize a restricted range of LiDAR sensors and platforms. Building on this progress, the SubT-MRS dataset (Zhao et al., 2024) by Team Explorer introduces additional scenes with varying degrees of degradation and weather changes, collected by multi-robot teams. Despite its contributions, this dataset is limited to a single spinning LiDAR, creating challenges for algorithms seeking compatibility with diverse hardware configurations. Ultimately, only algorithms that can maintain stable localization across heterogeneous datasets can truly be considered robust in a broader sense.

2.2. Datasets with heterogeneous LiDAR sensors

Heterogeneous LiDARs exhibit variances in resolution (high or low), FOV(large or narrow), scanning patterns (repetitive or non-repetitive), measurement noise, and the presence of additional channels such as intensity or reflectivity. These differences significantly impact LiDAR-based perception. For SLAM tasks, the Pohang Canal dataset (Chung et al., 2023) and the NTU VIRAL dataset (Nguyen et al., 2022) employ multiple spinning LiDARs to expand the perception area. The TIERS dataset (Qingqing et al., 2022) further incorporates non-repetitive LiDAR sensors with different scanning modes for multi-LiDAR SLAM. Similarly, the city dataset (Jung et al., 2023a) has been used to evaluate LiDAR SLAM with three heterogeneous LiDAR sensors. Beyond SLAM tasks, heterogeneous LiDAR data is utilized for other applications to explore the impact of sensor data characteristics on various tasks. For instance, the HeLiPR dataset (Jung et al., 2023b), which includes various types of LiDARs with different scanning patterns and additional measurement channels, is designed for place recognition with heterogeneous LiDARs, capturing spatiotemporal variations. The HeLiMOS dataset (Lim et al., 2024) focuses on moving object segmentation in 3D point clouds from heterogeneous LiDAR sensors. Furthermore, a Multi-LiDAR Multi-UAV Dataset (Catalano et al., 2023) was proposed to advance UAV tracking techniques. Despite the use of heterogeneous LiDARs in these datasets, their primary focus remains on SLAM in well-structured environments, place recognition, or object detection and tracking. There is still a need for datasets that address degradation detection and precise localization using heterogeneous LiDARs in degraded environments.

Compared to previous dataset works, our dataset exhibits enhanced comprehensiveness in four key aspects: (1) Extensive sensory measurements derived from diverse LiDARs, providing additional channels for LiDAR data; (2) Inclusion of multiple scenarios encompassing varying levels of degradation, enhancing the performance of degeneracy detection and mitigation; (3) Comprehensive data collection incorporating a wide range of motion patterns to facilitate algorithm design for general-purpose applications; and (4) Simulation of potential real-world sensor failures to improve adaptive algorithm switching and failure detection and recovery.

3. System overview

3.1. Sensors setup

The dataset's design aims to facilitate the development of LiDAR SLAM algorithms, independent of scanning modalities and FoV characteristics. To achieve this objective, we have developed three acquisition devices that share a common IMU and stereo camera but are equipped with distinct LiDAR sensors. The sensor parameters and layout are detailed in Table 2 and Figure 2(a). Our versatile acquisition system can be easily mounted on various platforms, as demonstrated in Figure 2(b)–(d), showcasing its adaptability to a handheld, sailboat, and UGV device, respectively.

3.2. Time synchronization scheme

Our FPGA-based synchronization module facilitates multi-channel sensor synchronization, as illustrated in Figure 3. The module is capable of achieving outdoor time synchronization through the reception of GNSS signals during initialization. By utilizing TIME_OF_DAY (TOD) and PPS signals from the GNSS, it generates synchronized signals at frequencies of 1, 10, and 100 Hz for LiDAR, cameras, and the Xsens MTi-30 IMU, respectively. The embedded IMUs in the Livox Avia and Ouster LiDARs autonomously coordinate with other sensors. In environments where GPS signal is unavailable, accurate synchronization is maintained using an internal clock mechanism. Although sensor trigger signals are generated simultaneously, unavoidable delays in signal transmission, data acquisition, and host communication result in minor timestamp differences. In our tests, the IMU–camera synchronization exhibited an error of approximately 4 ms, which is considered acceptable.

3.3. Sensor calibration

3.3.1. IMU calibration. We utilized the imu_utils[1] toolbox to perform calibration on the random walk and Gaussian white noise parameters of the Xsens MTi 30 IMU across the three devices. Furthermore, we employed LI-Init (Zhu et al., 2022) for bias calibration of the built-in IMUs in both the Livox and Ouster LiDAR sensors.

3.3.2. Stereo cameras calibration. We captured images of a 12 × 9 checkerboard using the sensor suite. The camera intrinsics and stereo extrinsic parameters were calibrated using a MATLAB toolbox, employing a pinhole camera model and a radial-tangential distortion model. The calibration results were obtained by minimizing the reprojection error and eliminating outliers with significant errors.

3.3.3. Camera-IMU extrinsic calibration. Utilizing the intrinsics of cameras and IMU, we employ the Kalibr[2] toolbox to perform spatial calibration of the IMU concerning all cameras. The process necessitated capturing the sensor suite's 6 degrees of freedom motion to ensure accuracy in the calibration results.

3.3.4. Camera-LiDAR extrinsic calibration. We employ MATLAB's LiDAR-Camera-Calibrator to calibrate Extrinsic between the mechanical LiDAR and the camera for devices α and β. For device γ, we utilize joint-LiDAR-camera-calib (Liu et al., 2022) to calibrate Livox Avia and the camera without checkerboards.

3.3.5. LiDAR-IMU extrinsic calibration. We performed external calibration of the LiDAR and Xsens MTi 30 IMU for the three devices using Li-Init (Zhu et al., 2022). Additionally, we conducted external parametric calibration for the Livox Avia and Ouster sensors along with their built-in IMUs.

3.3.6. Multi-LiDAR extrinsic calibration. For the Off-road and Inland Waterways sequences, where data from three devices were collected simultaneously, we performed LiDAR extrinsic calibration for the three devices using the MLCC method (Liu et al., 2022).

4. Dataset

The dataset encompasses a diverse range of geometrically degenerate scenarios, including flat surfaces, stairs, subway tunnels, off-road terrain, inland waterways, urban tunnels, and bridges. These scenarios present different levels and types of geometric degeneration, both translational and rotational. It includes data from various LiDAR sensors and platforms operating in challenging environments. The dataset consists of 64 sequences designed for algorithm development and evaluation, summarized in Table 3. These sequences feature both unidirectional forward trajectories and looping trajectories. Additional details about the trajectories can be found on the dataset's homepage. Figure 4 offers a visual overview of the scenarios.

4.1. Scenarios and challenges

4.1.1. Flat surfaces. Non-repetitive LiDARs with limited FoV face challenges when perceiving a single plane, such as walls or the ground. This leads to geometric degeneration in

Table 2. Sensor specifications and tracking devices.

Sensor type	Type	*Unit*
LiDAR α	Velodyne	
Model	VLP-16	
Scan lines	16	
Range	120	m
Vertical FOV	30	deg
Horizontal FOV	360	deg
Frequency	10	Hz
LiDAR β	Ouster	
Model	OS1-64	
Scan lines	64	
Range	100	m
Vertical FOV	45	deg
Horizontal FOV	360	deg
Frequency	10	Hz
IMU	InvenSense ICM-20948@100 Hz	
LiDAR γ	Livox	
Model	AVIA	
Range	450	m
Vertical FOV	30	deg
Horizontal FOV	360	deg
Frequency	10	Hz
IMU	BMI088@200 Hz	
IMU	Xsens	
Model	MTi-30 AHRS	
Frequency	100	Hz
Gyro noise density	0.03	$^{o}/s/\sqrt{Hz}$
Accel noise density	60	$\mu g/\sqrt{Hz}$
Mag RMS noise	0.5	mGauss
Camera—stereo	HikRobot - GigE cameras	
Model	MV-CS050-10GC	
Resolution	1224 × 1024	Pixel
Frequency	10	Hz
Readout method	Global shutter	
GNSS-RTK/INS	CHCNAV	
Mode	CGI610	
Position output	NMEA	
Frequency	100	Hz
RTK accuracy	1	cm
Mocap system	Vicon	
Mode	Vero 2.2	
Max frame rate	330	Hz
Accuracy	1	mm
Laser tracker	Leica	
Mode	Nova MS60	
Frequency	10	Hz
Accuracy	1	mm
3D laser scanner	Leica	
Mode	RTC360	
Range	130	m
Vertical FOV	300	deg
Horizontal FOV	360	deg
Accuracy	1	mm

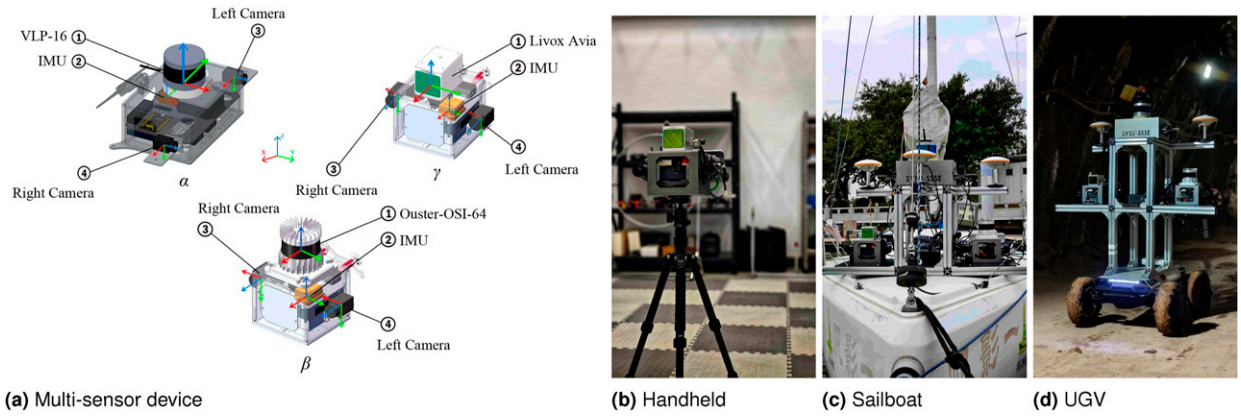


Figure 2. Multi-sensor devices and data collection platforms. (a) SolidWorks models of our sensor rig on three data collection devices, with the coordinate axes color-coded: red for the *X*-axis, green for the *Y*-axis, and blue for the *Z*-axis. This representation illustrates the transformation of sensor coordinates for each device. The multi-sensor rig mounted on (b) a handheld platform, (c) a sailboat, and (d) an UGV. The images in (b) through (d) demonstrate the diverse range of the GEODE dataset across various data collection platforms. (a) Multi-sensor device. (b) Handheld. (c) Sailboat. (d) UGV.

two translational dimensions (along the plane) and one rotational dimension (across the plane). In this scenario, we employed the Livox Avia LiDAR facing the ground with detailed textures, capturing sequences with both slow and aggressive motions.

4.1.2. Stairs. Stairs are confined spaces characterized by parallel planes (steps) and perpendicular planes (walls). The limited vertical FoV of sensors diminishes the information content, making scans less distinguishable. We utilized three devices with distinct scan lines and varying FoVs to gather data from corridors and stairs across multiple floors. Each device followed a path descending one of the stairs from the seventh floor and returning via the other one, creating a loop.

4.1.3. Metro tunnels. Subway tunnels exhibit geometric degeneration in both translational and rotational dimensions along the axial direction. Data collection within these tunnels utilized two construction methods: the shield method and the mine tunneling method. With the shield method, tunnel walls exhibited smooth surfaces, representing a highly degenerate scenario. In contrast, the mine tunneling method introduced irregular wall shapes, reducing the degree of geometric degeneration. A loop was formed by traversing several hundred meters along the tunnel and then doubling back. Some sequences were collected using multiple LiDAR sensors simultaneously to capture more spatial features.

4.1.4. Inland waterways. Water absorption reduces laser points, creating sparsely distributed features that weakly constrain the point cloud along the *Z*-axis and river flow, causing drift. A sudden increase in river width further reduces or eliminates valid LiDAR points, simulating a complete degradation scenario akin to LiDAR failure. Water surface reflections also disrupt signals, producing noisy or inaccurate measurements, which amplify noise-induced misalignment and affect position estimation in degraded directions. Sequences include one-way forward, forward-back, and repeated forward-back paths, with all heterogeneous LiDAR data collected simultaneously.

4.1.5. Off-road. A UGV was utilized as a data collection platform on an extensive grass track, with three devices deployed to capture data during both slow and vigorous movements. The off-road scenario demonstrates similar geometric degeneration to flat surfaces; however, long-distance traversal imposes greater demands on the algorithm to overcome such degeneration. The UGV completed a loop by returning to the starting point after collecting respective trajectory. All heterogeneous LiDAR data were collected simultaneously in this scenario.

4.1.6. Urban tunnel. Scan matching in urban tunnels is challenging due to the absence of distinct features or landmarks. These environments exhibit geometric degeneration, exacerbated by the presence of highly dynamic moving vehicles, significantly affecting point cloud registration. Furthermore, the low-texture surroundings within the tunnel and abrupt light variations during entry and exit impede camera effectiveness in compensating for LiDAR degeneracy.

4.1.7. Bridge. The bridge is flanked solely by the river on both sides, lacking distinctive landmarks for localization. Bridge scenes display limited textural variation, and the feature data in these areas tends to be uniform and repetitive, contributing to the challenge of achieving precise positioning.

4.2. Key features

4.2.1. Multi-degenerate scenarios. The diversity of degenerate scenarios in the GEODE dataset is reflected in both

the types and degrees of degeneracy. Geometric degeneracy in SLAM can be categorized into three types: rotational, translational, and combined. The dataset includes single-axis translational degeneracy scenarios such as tunneling tunnels, shield tunnels, bridges, urban tunnels, and stairs. Among these, tunneling tunnels have lower degeneracy due to their unique construction, while stairs exhibit degeneracy levels based on sensor information, with more data reducing degeneracy. Bridges and urban tunnels show higher degeneracy due to repetitive structures, which, despite offering some directional information, are overwhelmed by noise. Shield tunnels have the highest degeneracy, with nearly identical structures along their length, except for slight curves providing limited information.

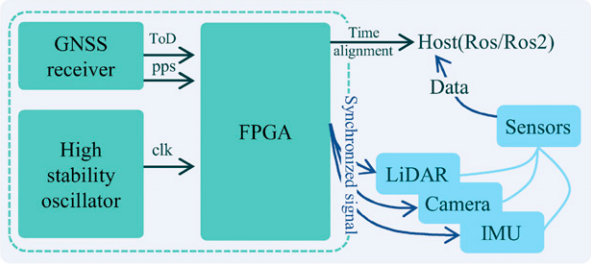


Figure 3. Time synchronization scheme.

Dual-axis translational degeneracy arises in flat ground, inland waterways, and off-road scenarios. In off-road environments, degeneracy primarily affects ground-parallel axes but is mitigated by sparse features like poles or shrubs. Inland waterways encounter Z-axis degeneracy and flow-aligned degradation, leading to elevation discrepancies in round-trip trajectories. Flat ground suffers total degeneracy along both ground-parallel axes, rendering LiDAR-only localization ineffective.

Rotational degeneracy appears in flat ground, shield tunnels, and off-road scenarios. Shield tunnels experience rotational degeneracy near the tunnel center, while off-road rotational degeneracy is milder. Flat ground suffers complete rotational degeneracy perpendicular to the ground. Inland waterways face six degrees of freedom degeneracy when sudden river widening causes LiDAR failures. GEODE's wide variety of these scenarios is crucial for preparing SLAM algorithms to handle practical challenges.

The dataset also includes environments across various scales, from small indoor spaces to medium-scale tunnels

Table 3. An overview of scenarios in GEODE dataset.

Scenario	Flat ground	Stairs	Metro tunnels	Off-road	Inland waterways	Urban tunnel	Bridges
Number of sequences	2	3	23	21	9	3	3
Size/GB	1.5	17.6	153.1	152.0	147.3	10.6	12.3
Duration/s	170	1066	6615	8112	7436	961	1174
Dist/m	108.29	901.81	7524.62	12829.20	15868.59	12975.32	14324.33
Degeneracy direction	2 trans. + 1 rot.	1 trans.	1 trans. + 1 rot.	2 trans. + 1 rot.	2 trans.	1 trans.	1 trans.
Devices (Platform)	γ (Handheld)	α (Handheld) β (Handheld) γ (Handheld)	α (UGV) β (UGV/ Handheld) γ (UGV/ Handheld)	α (UGV) β (UGV) γ (UGV)	α (Sailboat) β (Sailboat) γ (Sailboat)	α (Vehicle)	α (Vehicle)
Loop closure	✓	✓	✓	✓	✓	×	×
GT	Mocap	PALoc	Laser tracker	GNSS-RTK/ INS	GNSS-RTK/ INS	GNSS-RTK/ INS	GNSS-RTK/ INS

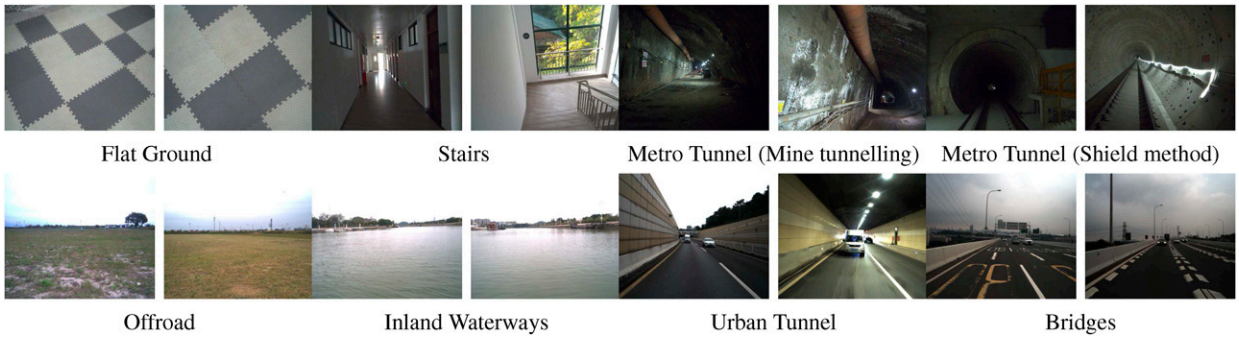


Figure 4. Images from different scenes.



Figure 5. Trajectories of multiple sequences captured in diverse outdoor scenarios, encompassing environments with varying scales and degrees of degradation, such as off-road areas, inland waterways, urban tunnels, and bridges.

and off-road tracks, and large-scale areas like waterways, bridges, and urban tunnels spanning tens of kilometers. Outdoor sequences, visualized in satellite views (e.g., Figure 5), highlight GEODE's extensive coverage. This breadth of degenerate scenarios and environments is vital for developing robust SLAM solutions capable of maintaining accuracy and reliability in diverse real-world applications.

4.2.2. Heterogeneous LiDAR. The degeneracy problem arises from insufficient information to effectively constrain pose optimization, influenced by environmental structure and sensor data can capture in such conditions. This creates a strong interplay between LiDAR types and geometric degeneracy. Heterogeneous LiDARs exhibit degeneracy patterns tied to scanning modes and noise distributions. Point clouds from different LiDARs in degraded environments vary in density, FOV, and noise characteristics, leading to unique degeneracy effects. Dense point clouds better exploit sparse features, reducing degeneracy. For instance, bushes on off-road sequences challenge low-beam spinning LiDARs more than non-repetitive LiDARs, which benefit from a longer range and denser scans. Similarly, expanding the horizontal FOV mitigates degeneracy on flat ground sequences and helps in scenarios like staircases by distinguishing repeated steps. Noise also exacerbates non-repetitive LiDARs produce more noise in rivers due to their range, while spinning LiDARs generate more uniform noise. Degeneracy mechanisms also impact algorithm robustness. Relying on a single LiDAR type risks overfitting to its characteristics, reducing adaptability. For example, in corridor degeneracy, spinning LiDARs struggle with "left-right symmetry," while non-repetitive LiDARs face challenges with sparse features and random scanning, lowering registration accuracy. Robust algorithms must ensure consistent performance across heterogeneous datasets to achieve generality.

The GEODE dataset tackles these challenges by incorporating heterogeneous LiDARs with diverse FOVs, scan lines, and additional information channels beyond geometry. This enables researchers to evaluate algorithms under various degenerate scenarios, assess the effects of LiDAR noise and reflection intensity on feature extraction and registration, and identify why certain algorithms succeed with specific sensors while failing with others. It also enables the integration of data from heterogeneous LiDARs to fill FoV gaps, boost point cloud density, and leverage additional channels to mitigate degeneracy.

4.2.3. Diverse platform characteristics. The GEODE dataset was collected across four unique platforms: a handheld device, an UGV, a sailboat, and a vehicle. Each platform exhibits unique motion characteristics, including speed, angular velocity, and dynamic frequency, as well as varying operational ranges. As depicted in Figure 6, the time-domain data captures the transient dynamics of each platform, while the frequency-domain data accentuates the salient features of their motion signatures. The handheld device is characterized by swift fluctuations in velocity and directional shifts. The UGV demonstrates motion profiles marked by high-frequency oscillations and abrupt jolts, attributable to the undulating terrain it navigates. In contrast, the sailboat's motion is governed by the ebb and flow of water currents, resulting in a syncopated mix of low-frequency drifts interspersed with high-frequency oscillations. The vehicular platform contributes data indicative of extensive, rapid transits, occasionally punctuated by sudden variations in pace and trajectory due to the exigencies of traffic and road conditions. This diverse dataset necessitates the development of robust SLAM algorithms capable of

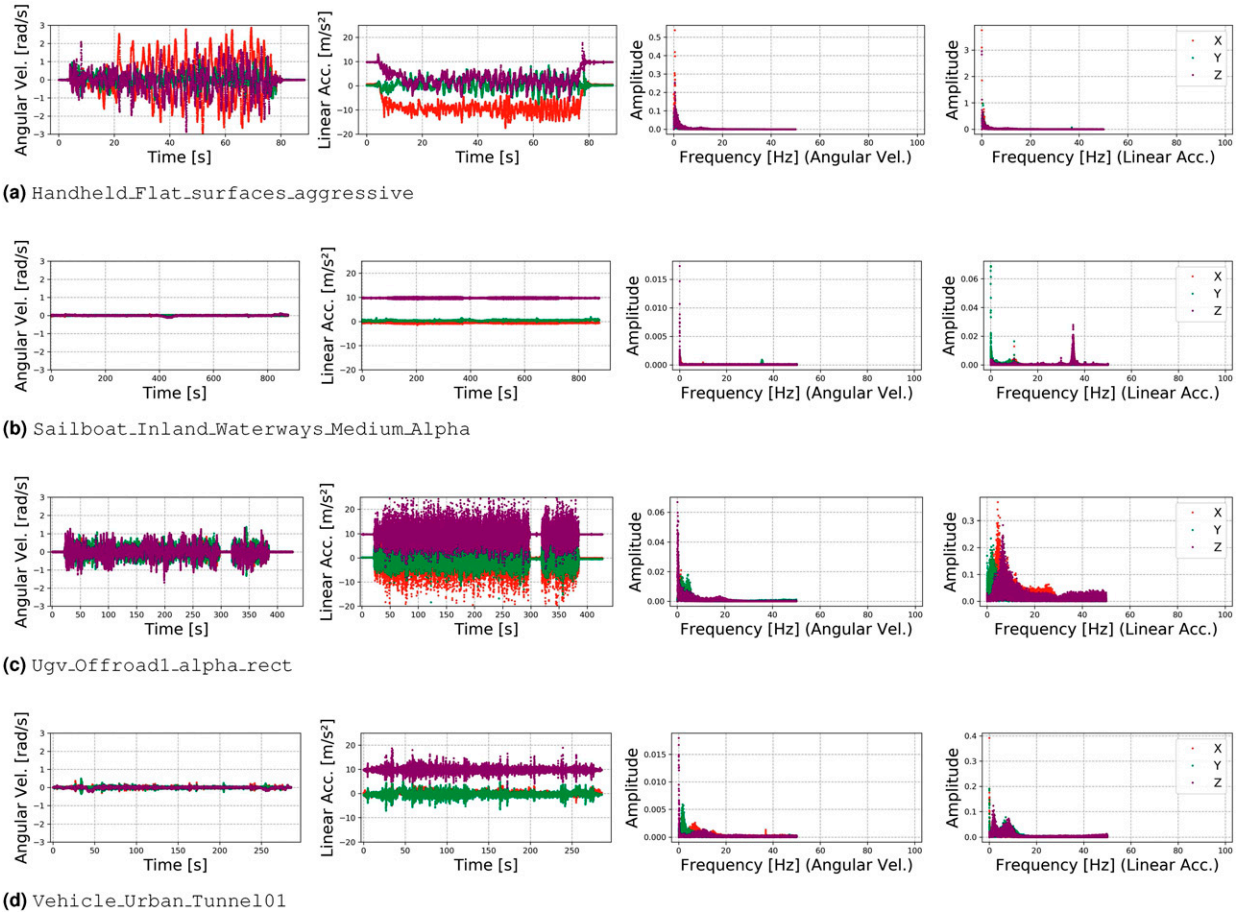


Figure 6. Cross-platform dynamic behavior and motion feature analysis in the GEODE dataset: insights from IMU data visualization. The motion characteristics are explored by leveraging visualization tools within the FusionPortableV2 Dataset, as referenced in Wei et al. (2024). (a) `Handheld_Flat_surfaces_aggressive`. (b) `Sailboat_Inland_Waterways_Medium_Alpha`. (c) `Ugv_Offroad1_alpha_rect`. (d) `Vehicle_Urban_Tunnel01`.

accommodating the varied motion profiles. The algorithms must be sensitive to the quick, erratic movements of handheld devices, the stable progression of UGVs, the oscillating patterns of sailboats, and the consistent velocities of vehicles. The GEODE dataset's comprehensive capture of these dynamics is crucial for creating SLAM solutions that are adaptable and reliable in a wide range of real-world scenarios.

4.3. Dataset organization

Figure 7 illustrates the structure of our dataset. Sensor data were collected using the Robot Operating System (ROS), with ROS2 (Macenski et al., 2022) for the α device and ROS1 (Quigley, 2009) for other devices. To accommodate users who do not use ROS, we provide the bag files alongside raw data in various formats: human-readable text, JPEG images, and binary LiDAR point cloud files. Additionally, we include tools for visualizing point cloud data from these binary files. The raw data has been compressed to enhance accessibility. We also supply calibration results for multiple sensors across three devices, a ground truth map for the stair sequence, and ground truth poses for all but one sequence. These resources ensure the dataset is comprehensive and useful for various applications and analyses.

4.3.1. Raw data formats. The raw data is organized and stored according to sensor type, with each category of sensor data housed in its respective folder. For IMU sensors, the recorded data is documented in a text file format (.txt), with each log entry on a separate line. Each line is prefixed by a timestamp, followed by the IMU sensor readings (roll, pitch, yaw, angular velocities, and linear accelerations). Camera data is stored separately for the left and right cameras, extracted from the compressed images within ROS bag files, and saved in JPEG format. These images are named based on the timestamps of their capture. LiDAR sensor data is stored in binary files (.bin), containing the range measurements for each frame. Velodyne data includes *x*, *y*, *z*, *intensity*, *ring*, *time*; Ouster data includes *x*, *y*, *z*, *intensity*, *t*, *reflectivity*, *ring*, *ambient*, *range*; and LiVOX data includes *x*, *y*, *z*, *intensity*, *tag*, *line*. These files contain essential data channels such as Cartesian coordinates, temporal offsets, and the indices of the rings or lines, crucial for data integrity and analysis.

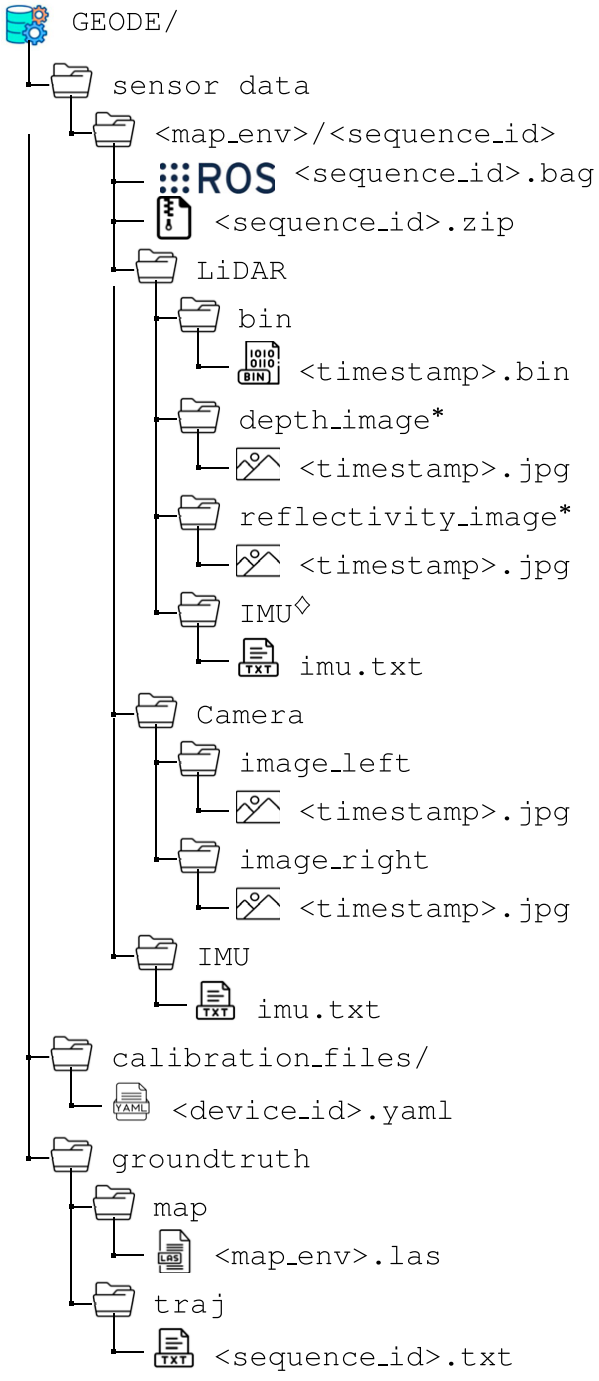


Figure 7. File structure of the GEODE dataset. *Depth and reflection images are exclusively accessible within the dataset when captured by the β device. Additionally, the built-in IMU data, denoted by $^{\diamond}$, is specifically available for the Ouster and Livox LiDAR.

4.3.2. ROS bag files. The ROS bag file format functions as an extensive repository, encompassing the complete sensor data and encapsulating all ROS topics originating from the respective distinct collection devices, as outlined in Table 4. This format guarantees the integration and preservation of the various streams of sensor information, facilitating seamless accessibility for ROS-based algorithms. For users of ROS2 bag, the rosbags-convert[3] toolkit can be used to convert formats between ROS1 and ROS2.

4.3.3. Calibration files. The outcomes of the multi-sensor calibration, as described in Section 3.3, are recorded in three YAML files, each corresponding to the respective acquisition devices.

4.4. Groundtruth generation

4.4.1. LiDAR point cloud maps generation. For the stairs sequence, we employed a Leica RTC360 laser scanner to meticulously capture the details of the stairs on both sides of a building and the corridors across multiple floors with millimeter accuracy.

4.4.2. Groundtruth poses. To capture accurate motion ground truth in various degraded environments of the GEODE dataset, it is essential to use appropriate equipment tailored to the specific scene characteristics, as illustrated in Figure 8(a). For outdoor sequences such as urban tunnels, bridges, inland waterways, and off-road settings, we use the RTK-INS device CHCNAV CG610 to obtain 6-DoF groundtruth poses. In the bridge and urban tunnel sequences, data collection relies solely on the α device. Hand-eye calibration (Furrer et al., 2017) is employed to refine the positioning results from the RTK-INS, ensuring accurate motion values. In the inland waterway and off-road sequences, three acquisition devices are mounted on a rack constructed from aluminum profiles, which is then attached to a sailboat or UGV for data acquisition. The RTK-INS device's trajectory data is synchronized with the β device using hand-eye calibration. Further processing with multi-LiDAR calibration (Liu et al., 2022) refines the motion values for the α and γ devices. Using the CHCNV CG610 with dual antennas and differential base station services, the nominal accuracy of the obtained ground truth trajectory can reach 1 cm. However, when fewer satellites are accessible, Z-axis errors become significantly larger than XY-axis errors, and false positive issues are also likely to occur. In the urban tunnel sequence, the trajectory accuracy error is particularly significant when the car enters and exits the tunnel. Therefore, we only use the fixed solutions as the ground truth trajectory and manually remove some points with large errors.

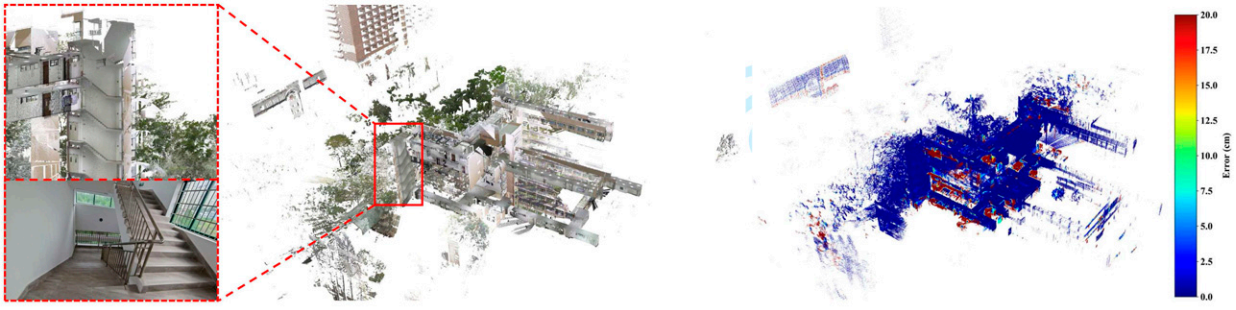
For sequences on flat surfaces, 6-DoF ground truth is captured using the Vicon motion capture system. The positioning accuracy of motion capture using Vicon is 1 mm. In metro tunnels, the Leica MS 60 is used to track prisms and obtain ground truth position. Temporal alignment between Leica MS60 total station measurements and sensor data is achieved using tool in FusionPortableV2 (Wei et al., 2024). Similar to the inland waterways sequence, some metro tunnel data is collected using three devices simultaneously. The trajectory data from the Leica MS60 is synchronized to the α device via hand-eye calibration, while the β and γ devices' ground truth is obtained through further

Table 4. Sensor data descriptions and specifications.

Device	Sensor	Message type	Topic name	Description	Frequency (Hz)
α	LiDAR	sensor_msgs/ PointCloud2	/velodyne_points	Raw velodyne pointcloud with [*x*, *y*, *z*, *intensity*, *ring*, *time*]	10
	Camera	sensor_msgs/ CompressedImage	/left_camera/compressed	Compressed RGB image	10
		sensor_msgs/ CompressedImage	/right_camera/ compressed	Compressed RGB image	10
	IMU	sensor_msgs/Imu	/imu/data	Raw IMU data from Xsens MTi-30	100
β	LiDAR	sensor_msgs/ PointCloud2	/ouster/points	Raw ouster pointcloud with [*x*, *y*, *z*, *intensity*, *t*, *reflectivity*, *ring*, *ambient*, *range*]	10
		sensor_msgs/Image	/ouster/reflec_image	Reflectivity data	10
		sensor_msgs/Image	/ouster/range_image	Depth image	10
		sensor_msgs/Imu	/ouster/imu	Raw IMU data from InvenSense ICM-20948	100
	Camera	sensor_msgs/ CompressedImage	/left_camera/image/ compressed	Compressed RGB image	10
		sensor_msgs/ CompressedImage	/right_camera/image/ compressed	Compressed RGB image	10
	IMU	sensor_msgs/Imu	/imu/data	Raw IMU data from Xsens MTi-30	100
γ	LiDAR	livox_ros_driver/ CustomMsg	/livox/lidar	Livox defined cloudpoint data with [*x*, *y*, *z*, *intensity*, *tag*, *line*]	10
		sensor_msgs/Imu	/livox/imu	Raw IMU data from BMI088	200
	Camera	sensor_msgs/ CompressedImage	/left_camera/image/ compressed	Compressed RGB image	10
		sensor_msgs/ CompressedImage	/right_camera/image/ compressed	Compressed RGB image	10
	IMU	sensor_msgs/Imu	/imu/data	Raw IMU data from Xsens MTi-30	100



(a) Devices for generating GT trajectories and maps: CHCNAV CG610, Vicon Vero 2.2, Leica Nova MS60, and Leica RTC360.



(b) GT RGB point cloud map of "Stairs" sequence. **(c)** Groundtruth evaluation.

Figure 8. Groundtruth Generation. (a) Devices for generating GT trajectories and maps: CHCNAV CG610, Vicon Vero 2.2, Leica Nova MS60, and Leica RTC360. (b) GT RGB point cloud map of "stairs" sequence. (c) Groundtruth evaluation.

multi-LiDAR calibration processing. The positioning accuracy of the laser tracker with a tracking prism is 1 mm.

For the stairs sequence, we provide a ground truth map obtained using the Leica RTC360, offering an accuracy of 1 mm. Additionally, we include ground truth trajectories generated via the PALoc SLAM method (Hu et al., 2024b), which estimates LiDAR poses within the pre-constructed ground truth map depicted in Figure 8(b). For the trajectories generated by PALoc, the inlier RMSE for the α sequence is 0.07 m, and for the β sequence, it is 0.08 m.

Figure 8(c) illustrates maps constructed from the ground truth poses generated by PALoc, highlighting errors from the ground truth map in the sequence "stair_bob." Due to the limited FOV of the Livox Avia LiDAR on the γ device, the PALoc algorithm could not generate an accurate ground truth trajectory, even with a ground truth map available. Consequently, for this sequence, we advise users of this dataset to compare the map produced by their algorithm with the provided ground truth map to assess localization errors using the MapEval toolkit (Hu et al., 2024a).

4.5. Development tools

We offer a comprehensive development kit[4] to assist users in effectively utilizing our dataset, particularly for evaluating algorithms implemented in Python. Given the limited three degrees of freedom of the ground truth pose captured by the Leica MS 60 tracking prism in a metro tunnel, transforming these poses into the body coordinate systems of both the β and γ devices using multi-LiDAR calibration results can be challenging. To facilitate this process, we provide Python scripts that convert trajectory results from users' algorithms to the α device's body coordinate system for localization evaluation. Similarly, for the RTK-INS-based ground truth pose, which has six degrees of freedom, we offer scripts for converting user-generated trajectories to the β device's coordinate system.

The development kit also includes scripts for calculating errors and visualizing trajectories, ensuring users have a comprehensive toolkit for algorithm assessment and validation. Furthermore, we offer a comprehensive suite of C++ code, developed utilizing the software toolkit referenced in Morales et al. (2021), designed to extract raw data from ROS bag files. Accompanying this, we also provide scripts tailored for visualizing LiDAR point clouds derived from binary data.

5. Evaluation

This section delivers an extensive examination of cutting-edge LiDAR-based odometry methodologies, utilizing the GEODE dataset as a benchmark. The thorough assessment is designed to delineate the weakness of contemporary LiDAR odometry techniques and underscore the pivotal role that datasets akin to GEODE play in propelling robust algorithms forward. Our evaluation was conducted on eight methods, including LIO, LIVO, and stereo VIO. For each sequence, we performed five tests and averaged the results to obtain the corresponding localization errors for reference.[5]

5.1. Baselines

FAST-LIO2 (Xu et al., 2021) offers a dependable solution for LiDAR-inertial odometry by fusing IMU data with LiDAR feature points through an iterative extended Kalman filter. It enables efficient navigation in challenging environments characterized by rapid motion, noise, or clutter, where performance degradation may occur.

DLIO (Chen et al., 2022) is a newly introduced odometry that construct continuous-time trajectories for precise motion correction. It introduces several enhancements over its predecessor, DLO (Chen et al., 2021), initially developed for the DARPA Challenge.

COIN-LIO (Pfreundschuh et al., 2024) proposes a pipeline leverage intensity as an additional modality to improve the robustness of LiDAR-inertial odometry in geometrically degenerate scenarios.

LVI-SAM (Shan et al., 2021) is a LiDAR-Inertial-Visual SLAM system with integrated mechanisms for LiDAR degeneration detection, odometry failure identification, and recovery.

R3LIVE (Lin and Zhang, 2021) is a fusion framework that combines LiDAR, inertial, and visual sensors to achieve reliable and precise state estimation in challenging environments with geometric degeneracy.

FAST-LIVO (Zheng et al., 2022) is a tightly coupled odometry system designed to achieve localization in environments lacking structure and texture by leveraging photometric error and point-to-plane residuals.

VINS-Fusion (Qin et al., 2018) is an optimization-based, multi-sensor state estimator that extends the capabilities of VINS-Mono. It supports stereo cameras combined with an IMU, enabling precise localization.

5.2. Accuracy evaluation

Table 5 summarizes the Absolute Trajectory Errors (ATE) observed in real-world geometrically degraded scenarios for eight SLAM algorithms.[6] The table does not present the results for the flat ground scenario, where only VINS-Fusion achieved localization errors of 1.10 m and 2.20 m in the smooth and aggressive sequences, respectively, while all other methods experienced drift. For the remaining sequences not listed, with the exception of FAST-LIVO which had a localization error of 64.68 m in the bridge01 sequence, all methods failed to produce meaningful trajectory outputs. The results highlight even though these SLAM systems perform exceptionally well in standard scenarios, there is still room for improvement in degraded environments. Three potential reasons for their failure can be attributed to the following factors.

5.2.1. Robust to geometrical degeneracy. In scenarios with mild to moderate degradation, there still exists relatively

Table 5. Evaluation of ATE (M) of SLAM systems on dataset sequences. We use colors to categorize the algorithms into LIO, LIVO, and VIO.

Sequence and degeneracy direction				COIN-LIO	FAST-LIO2	DLIO	FAST-LIVO	Coco-LIC	R3LIVE	LVI-SAM	VINS-fusion
Off-road	α	01	⊙↗	—	**0.18**	×	0.20	—	×	×	×
		02	⊙↗	—	**0.39**	×	0.50	—	×	×	×
		03	⊙↗	—	**0.33**	×	0.38	—	×	×	×
		04	⊙↗	—	**0.43**	×	1.51	—	×	×	×
		05	⊙↗	—	**0.30**	×	0.58	—	×	×	×
		06	⊙↗	—	**0.22**	×	0.29	—	×	×	×
		07	⊙↗	—	**0.26**	×	0.50	—	×	×	×
	β	01	⊙↗	**0.12**	**0.12**	0.18	0.13	1.44	5.42	0.14	×
		02	⊙↗	**0.13**	0.31	0.22	**0.13**	31.17	20.95	0.20	×
		03	⊙↗	**0.12**	0.34	0.18	0.14	23.20	7.68	2.76	×
		04	⊙↗	0.13	0.14	0.23	**0.09**	27.50	37.79	×	×
		05	⊙↗	**0.16**	0.19	0.37	0.18	39.01	42.43	8.33	56.10
		06	⊙↗	0.15	0.15	0.21	**0.13**	28.08	4.78	×	×
		07	⊙↗	**0.11**	**0.11**	0.15	**0.11**	11.21	3.77	19.19	×
	γ	01	⊙↗	—	0.09	1.19	**0.08**	0.45	×	2.43	×
		02	⊙↗	—	**0.11**	8.81	**0.11**	0.41	×	6.40	×
		03	⊙↗	—	**0.09**	0.29	0.13	0.41	×	1.14	×
		04	⊙↗	—	×	3.55	**0.24**	0.44	×	2.61	×
		05	⊙↗	—	**0.13**	17.58	0.15	0.41	×	1.60	×
		06	⊙↗	—	**0.12**	0.87	0.13	0.29	×	27.04	×
		07	⊙↗	—	**0.16**	5.71	0.19	0.32	5.65	1.11	×
Inland waterways	α	Short	↗	—	10.27	**2.59**	17.91	—	4.80	×	×
		Medium	↗	—	19.69	**13.25**	32.46	—	22.77	×	×
		Long	↗	—	70.26	68.04	**33.36**	—	×	×	×
	β	Short	↗	**0.13**	0.29	0.15	0.20	×	3.53	×	×
		Medium	↗	**0.49**	0.57	0.72	10.61	×	47.68	×	×
		Long	↗	**0.78**	0.99	0.82	19.42	×	×	×	×
	γ	Short	↗	—	0.24	0.39	**0.23**	×	0.35	23.81	×
		Medium	↗	—	1.82	2.75	**0.73**	×	5.73	×	×
		Long	↗	—	6.90	5.34	**0.37**	×	12.66	×	×
Metro tunnels	α	Tunneling 3	↗	—	0.21	**0.18**	0.20	—	0.59	×	47.66
		Tunneling 4	↗	—	0.24	**0.14**	0.19	—	0.25	×	×
		Tunneling 5	↗	—	0.19	**0.13**	0.17	—	11.44	0.30	0.69
	β	Tunneling 2	↗	0.14	0.14	**0.11**	0.12	0.15	0.14	0.14	2.31
		Tunneling 3	↗	0.23	0.17	**0.14**	0.15	0.19	0.20	0.37	×
		Tunneling 4	↗	0.16	0.17	**0.15**	0.27	0.17	0.25	0.18	×
		Tunneling 5	↗	3.84	0.12	**0.11**	0.13	0.18	0.26	0.22	0.36
	γ	Tunneling 1	↗	—	1.16	8.51	**0.36**	0.42	2.83	35.40	×
		Tunneling 2	↗	—	1.88	15.43	**1.56**	2.06	86.51	2.10	×
		Tunneling 3	↗	—	**0.19**	2.63	0.20	4.06	63.05	0.24	×
		Tunneling 4	↗	—	0.15	5.74	**0.13**	0.18	57.46	0.35	×
		Tunneling 5	↗	—	0.27	2.48	**0.12**	0.17	1.30	0.17	22.03
Stairs	α		↗	—	4.69	4.89	**3.28**	—	4.54	×	3.66
	β		↗	0.45	**0.38**	0.41	0.40	6.84	×	×	0.59
	γ		↗	—	×	×	×	×	×	×	×

[a]⊙ indicates that the sequence contains scenarios with rotational degeneracy, while ↗ signifies that the scenario includes translational degeneracy.
[b]× denotes the algorithm breaking down or having a localization error larger than 100 m. — denotes the SLAM algorithm is not adapted to this data.

high-distinctiveness geometric information; however, many methods may still fail in these situations. This could be related to how point cloud data is handled during registration. In geometrically degraded scenarios, the features useful for localization are often sparse. Feature extraction approaches may result in the loss of substantial geometric information, leading to a decline in performance. On the other hand, using dense methods that register the entire point cloud may cause the sparse features to be overwhelmed by repetitive yet low-information point cloud data. Furthermore, the impact of degraded environments varies for different algorithms depending on the type of LiDAR used. For instance, as indicated by the results from the Inland Waterways scenario, localization using data collected with non-repetitive LiDARs often outperforms that using sparse spinning LiDAR data. Specifically, for the Fast-LIVO algorithm, non-repetitive LiDAR data achieves the highest localization accuracy. However, in the Stairs scenario, spinning LiDARs with a larger field of view tend to enable more accurate localization, while non-repetitive LiDAR data results in significant drift. Additionally, COIN-LIO, which leverages intensity and geometric structural information, performs well in scenarios with mild degradation but still fails as the degree of degradation increases. From the results of stereo VIO, it is evident that relying solely on visual information leads to failure in most scenarios. However, in specific scenarios where LiDAR is completely ineffective, such as Flat Ground, stereo VIO achieves favorable performance. This highlights a potential pathway for improvement: in addition to addressing LiDAR data degeneracy, incorporating visual information could be a viable approach.

5.2.2. Without adaptive capabilities. A significant limitation of current SLAM methods is their inability to actively select the most informative measurements to adapt to various environments. This issue is evident in two main aspects. Firstly, single-sensor odometry lacks the capability for adaptive feature tracking. For example, all LIO methods failed to utilize the steel cables of the Cable Stayed Bridge and street lights as key localization features in the bridge sequence. Consequently, points that could be accurately located in the degraded direction (represented by orange points in Figure 9(a), as assessed by the RMS algorithm (Petracek et al., 2024)) are overshadowed by repetitive, non-contributing points (blue dots in Figure 9(a)), leading to significant drift. Similarly, the VIO module in multi-sensor fusion methods such as LVI-SAM and R3LIVE does not adequately reduce the weight of feature points on water surfaces in inland waterway sequences, resulting in visual localization drift, as depicted in Figure 9. Secondly, multi-sensor fusion algorithms often fail to dynamically select complementary sensors to maintain uninterrupted localization. In tunnel sequences, which are characterized by varying exposure levels and featureless surroundings, algorithms designed to combine LiDAR and visual sensors should leverage visual data in geometrically degraded contexts while ignoring it under over-exposure conditions. Similarly, environments such as inland waterways and flat ground sequences present challenges for algorithms to quickly identify sensor degeneration and switch to alternative modalities for localization. However, current multi-sensor fusion algorithms have consistently underperformed in such scenarios, highlighting their limitations.



Figure 9. Lack of adaptive feature tracking capability. (a) Point cloud sampling. (b) Visual feature point tracking.

5.2.3. Lack of failure detection and recovery. During testing, many algorithms exhibited trajectory divergence under the influence of degradation, which can be attributed to the lack of adaptive architectures to effectively handle algorithm failures. This deficiency is primarily reflected in two aspects: the inability to accurately detect sensor failure states and the absence of mechanisms to recover from such failures, resulting in significant challenges when dealing with real-world degradation. Currently, most algorithms are unable to correctly identify the direction of degradation in point cloud alignment, relying instead on auxiliary state estimation that is highly susceptible to noise. Similarly, they lack effective outlier measurement rejection, leading to failures in self-similar areas (affecting LiDAR) and under poor lighting conditions (affecting vision). Consequently, many LIVO algorithms perform even worse than LIO algorithms in these cases. Moreover, current odometry algorithms lack the capability to recover localization after submodule failures. For instance, despite LVI-SAM's claim of utilizing a multi-sensor subsystem to reinitialize failed subsystems, it failed to reorient the LiDAR-inertial odometry after experiencing geometric degradation during testing. These findings highlight the urgent need for more robust and adaptive algorithms capable of handling sensor failures and geometric degradation in diverse and challenging environments.

5.3. Future research directions

To achieve accurate odometry estimates in geometrically degenerated scenarios, resilient methods capable of dynamic reconfiguration are required. This involves several aspects, such as selecting residuals, adding constraints (Tuna et al., 2024a, 2024b), online adaptive tuning of parameters, and even switching algorithmic modules. For example, treating the residuals in different degrees of

freedom independently for LIO methods can improve applicability by reducing the need for domain-specific tuning. Moreover, dynamically utilizing information from diverse LiDAR systems can improve perception in challenging environments, offering potential solutions to localization difficulties.

For visual odometry that assists LiDAR odometry, challenges such as texture repetition, lighting variations, and occlusion must be addressed, and outlying points should be automatically rejected (Bai et al., 2021). In multi-sensor fusion, it is crucial to assign appropriate weights to different measurements to enhance the system's robustness (Chen et al., 2024b). Determining the reliability of each sensor source in various directions is essential in a multi-source fusion system, highlighting the significant research value in recognizing degradation scenarios.

Another important area of research is how to recover the algorithmic module after a failure. Extending the robustness of LiDAR SLAM to geometrically degenerate scenarios requires redundancy and resourcefulness in information to constrain state optimization in the degenerated direction. This should be achieved through means beyond merely adding more sensors. It is necessary to regain performance in the face of degraded sensing or environmental changes.

6. Known issues

Despite our meticulous efforts to construct a multifaceted dataset encompassing diverse platforms, sensors, and scenarios, the process is inherently challenging and not without imperfections. We acknowledge the limitations in our data processing and the persistent challenges despite our diligent data collection and curation efforts. In the following sections, we outline these prevalent issues and propose technical mitigations, hoping to offer future researchers valuable insights and instructive lessons.

6.1. Camera exposure setting

We set the camera exposure time to a fixed value to reduce image differences caused by lighting changes, making the dataset more uniform. However, this introduces limitations, such as the inability to obtain optimal image quality in extremely bright or dark environments. This issue is particularly evident in underground scenes with insufficient lighting, leading to underexposed images with limited visibility, which poses challenges for most visual perception algorithms.

6.2. Calibration among multiple sensors

Although each collection device is calibrated using a robust calibration scheme and mounted on rigid aluminum-alloy-based parts to minimize external interference, the calibration process is conducted only once per machine. For data collection efforts spanning a week, the calibration results may not maintain high precision for specific iterations. We recommend users start with our calibration results and investigate new methods for online calibration. We have also made the calibration data available on the dataset's webpage, allowing the community to perform more precise calibrations.

6.3. Time synchronization

Our synchronization technique ensures that the LiDAR, cameras, and IMU are all activated by the same clock. However, the computer assigns timestamps to each sensor's data differently due to varying transmission latencies and decoding times. The LiDAR directly receives GNSS time as its clock source, so delays caused by data transmission and processing are negligible. However, the IMU and camera do not have direct access to GNSS time. For the IMU, we disregard minor transmission and processing delays, synchronizing GNSS time with the computer NUC used for acquisition and running the ROS driver to time the messages. For the camera, we use shared memory to store the latest time and estimate the exposure time to obtain image timestamps. Despite these measures, ensuring perfect synchronization of all sensor data remains challenging. Users requiring higher timestamp accuracy can estimate the time differences between different sensors online.

7. Conclusion and future work

This study introduces GEODE, a meticulously crafted dataset designed to improve robustness in geometrically degenerate environments. The dataset features three data collection devices, each equipped with different types of LiDARs, as well as the same stereo camera and IMU, all of which have been carefully calibrated and synchronized. These devices are installed on various platforms, collecting data across seven different degenerate scenarios: flat surfaces, stairs, subway tunnels, off-road terrain, inland waterways, urban tunnels, and bridges. GEODE provides a valuable benchmark for LiDAR-based odometry in previously underrepresented scenarios, encouraging a shift in research focus from achieving higher accuracies in geometrically simple cases to enhancing robustness in more complex environments.

Looking forward, we aim to expand GEODE by incorporating additional challenging scenarios and continuously enhancing its quality. We also plan to maintain a benchmark with undisclosed ground truth to facilitate more objective and meaningful comparisons of algorithm performance in degraded environments. Through these efforts, we hope to further improve the robustness and reliability of robotic systems, ensuring their effectiveness across a broader range of real-world applications.

Acknowledgments

The authors gratefully acknowledge Shipeng Zhong, Ruilan Yang, Qiming Chen, and Yizhen Yin for their contributions to sensor calibration and ground truth pose processing. We also thank the organizational staff and participants for their involvement in the exercises.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

Zhiqiang Chen https://orcid.org/0000-0001-9011-4981
Xiangcheng Hu https://orcid.org/0000-0003-3535-6886

Notes

1. imu_utils: https://github.com/gaowenliang/imu_utils.
2. Kalibr: https://github.com/ethz-asl/kalibr.
3. rosbags-convert: https://ternaris.gitlab.io/rosbags/.
4. Development Tools: https://thisparticle.github.io/geode/.
5. We did not fine-tune the parameters for each sequence to achieve the potentially optimal performance of each algorithm; thus, the reported results are provided for reference only. Detailed experimental setups can be found at https://github.com/thisparticle/GEODE_Evaluation.
6. A more comprehensive table, along with the inclusion of subsequent tests for new methods, can be found at https://thisparticle.github.io/geode/\#OnlineBenchmark.

References

Bai X, Wen WW and Hsu L (2021) Degeneration-aware outlier mitigation for visual inertial integrated navigation system in urban canyons. *IEEE Transactions on Instrumentation and Measurement* 70: 1–15.

Carlevaris-Bianco N, Ushani AK and Eustice RM (2016) University of Michigan north campus long-term vision and lidar dataset. *The International Journal of Robotics Research* 35(9): 1023–1035.

Catalano I, Yu X and Queralta JP (2023) Towards robust UAV tracking in GNSS-denied environments: a multi-lidar multi-UAV dataset. In: 2023 IEEE international conference on robotics and biomimetics (ROBIO), Koh Samui, Thailand, 4–9 December 2023, 1–7. IEEE.

Chang Y, Ebadi K, Denniston C, et al. (2022) Lamp 2.0: a robust multi-robot slam system for operation in challenging large-scale underground environments. *IEEE Robotics and Automation Letters* 7: 9175–9182.

Chen K, Lopez BT, Akbar Agha-mohammadi A, et al. (2021) Direct lidar odometry: fast localization with dense point clouds. *IEEE Robotics and Automation Letters* 7: 2000–2007.

Chen K, Nemiroff R and Lopez BT (2022) Direct lidar-inertial odometry: lightweight lio with continuous-time motion correction. In: 2023 IEEE international conference on robotics and automation (ICRA), London, UK, 29 May–2 June 2023, 3983–3989.

Chen P, Guan W, Huang F, et al. (2024a) Ecmd: an event-centric multisensory driving dataset for slam. *IEEE Transactions on Intelligent Vehicles* 9(1): 407–416.

Chen Z, Chen H, Qi Y, et al. (2024b) Relead: resilient localization with enhanced lidar odometry in adverse environments. In: 2024 IEEE international conference on robotics and automation (ICRA), Yokohama, Japan, 13–17 May 2024, 3999–4005.

Chung D, Kim J, Lee C, et al. (2023) Pohang canal dataset: a multimodal maritime dataset for autonomous navigation in restricted waters. *The International Journal of Robotics Research* 42(12): 1104–1114.

Ebadi K, Bernreiter L, Biggie H, et al. (2024) Present and future of slam in extreme environments: the darpa subt challenge. *IEEE Transactions on Robotics* 40: 936–959.

Feng D, Qi Y, Zhong S, et al. (2024) S3e: A multi-robot multimodal dataset for collaborative slam. *IEEE Robotics and Automation Letters* 9(12): 11401–11408.

Furrer F, Fehr M, Novkovic T, et al. (2017) Evaluation of combined time-offset estimation and hand-eye calibration on robotic datasets. In: International symposium on field and service robotics, Zurich, Switzerland, 12–15 September 2017.

Geiger A, Lenz P and Urtasun R (2012) Are we ready for autonomous driving? The kitti vision benchmark suite. In: 2012 IEEE conference on computer vision and pattern recognition, Providence, RI, 16–21 June 2012, 3354–3361.

Helmberger M, Morin K, Berner B, et al. (2021) The hilti slam challenge dataset. *IEEE Robotics and Automation Letters* 7: 1–8.

Hsu L, Huang F, Ng HF, et al. (2023) Hong Kong urbannav: an open-source multisensory dataset for benchmarking urban navigation algorithms. *Navigation: Journal of the Institute of Navigation* 70(4): navi.602.

Hu X, Wu J, Jia M, et al. (2024a) Mapeval: towards unified, robust and efficient slam map evaluation framework. ArXiv Preprint arXiv:2411.17928.

Hu X, Zheng L, Wu J, et al. (2024b) Paloc: advancing slam benchmarking with prior-assisted 6-dof trajectory generation and uncertainty estimation. *IEEE* 29: 1–12.

Jung M, Jung S and Kim A (2023a) Asynchronous multiple lidar-inertial odometry using point-wise inter-lidar uncertainty propagation. *IEEE Robotics and Automation Letters* 8(7): 4211–4218.

Jung M, Yang W, Lee D, et al. (2023b) Helipr: heterogeneous lidar dataset for inter-lidar place recognition under spatiotemporal variations. *The International Journal of Robotics Research* 43: 02783649241242136.

Li J, Wu W, Yang B, et al. (2023) Whu-helmet: a helmet-based multisensor slam dataset for the evaluation of real-time 3-d mapping in large-scale gnss-denied environments. *IEEE Transactions on Geoscience and Remote Sensing* 61: 1–16.

Liao Y, Xie J and Geiger A (2021) Kitti-360: a novel dataset and benchmarks for urban scene understanding in 2d and 3d. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45: 3292–3310.

Lim H, Jang S, Mersch B, et al. (2024) Helimos: a dataset for moving object segmentation in 3d point clouds from heterogeneous lidar sensors. ArXiv Preprint arXiv:2408.06328.

Lin J and Zhang F (2021) R3live: a robust, real-time, rgb-colored, lidar-inertial-visual tightly-coupled state estimation and mapping package. In: 2022 international conference on robotics and automation (ICRA), Philadelphia, PA, 23–27 May 2022, 10672–10678.

Liu X, Yuan C and Zhang F (2022) Targetless extrinsic calibration of multiple small fov lidars and cameras using adaptive voxelization. *IEEE Transactions on Instrumentation and Measurement* 71: 1–12.

Macenski S, Foote T, Gerkey BP, et al. (2022) Robot operating system 2: design, architecture, and uses in the wild. *Science Robotics* 7: 6074.

Morales J, Vázquez-Martín R, Mandow A, et al. (2021) The umasar dataset: multimodal data collection from a ground vehicle during outdoor disaster response training exercises. *The International Journal of Robotics Research* 40: 835–847.

Nguyen TM, Yuan S, Cao M, et al. (2022) Ntu viral: a visual-inertial-ranging-lidar dataset, from an aerial vehicle viewpoint. *The International Journal of Robotics Research* 41(3): 270–280.

Nguyen TM, Yuan S, Nguyen TH, et al. (2024) Mcd: diverse large-scale multi-campus dataset for robot perception. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, Seattle, WA, 16–22 June 2024, pp. 22304–22313.

Petracek P, Alexis K and Saska M (2024) Rms: redundancy-minimizing point cloud sampling for real-time pose estimation. *IEEE Robotics and Automation Letters* 9(6): 5230–5237. DOI: 10.1109/LRA.2024.3389820.

Petravcek P, Kratky V, Petrlik M, et al. (2021) Large-scale exploration of cave environments by unmanned aerial vehicles. *IEEE Robotics and Automation Letters* 6: 7596–7603.

Pfreundschuh P, Oleynikova H, Cadena C, et al. (2024) Coin-lio: complementary intensity-augmented lidar inertial odometry. In: 2024 IEEE international conference on robotics and automation (ICRA), Yokohama, Japan, 13–17 May 2024, 1730–1737. IEEE.

Qin T, Li P and Shen S (2018) Vins-mono: a robust and versatile monocular visual-inertial state estimator. *IEEE Transactions on Robotics* 34(4): 1004–1020.

Qingqing L, Xianjia Y, Queralta JP, et al. (2022) Multi-modal lidar dataset for benchmarking general-purpose localization and mapping algorithms. In: 2022 IEEE/RSJ international conference on intelligent robots and systems (IROS), Kyoto, Japan, 23–27 October 2022, 3837–3844. IEEE.

Quigley M (2009) Ros: an open-source robot operating system. In: IEEE international conference on robotics and automation, Kobe, Japan, 12–17 May 2009.

Shan T, Englot B, Ratti C, et al. (2021) Lvi-sam: tightly-coupled lidar-visual-inertial odometry via smoothing and mapping. In: 2021 IEEE international conference on robotics and automation (ICRA), Xian, China, 30 May 2021–5 June 2021, 5692–5698.

Tranzatto M, Miki T, Dharmadhikari M, et al. (2022) Cerberus in the darpa subterranean challenge. *Science Robotics* 7: eabp9742.

Tuna T, Nubert J, Nava Y, et al. (2024a) X-icp: localizability-aware lidar registration for robust localization in extreme environments. *IEEE Transactions on Robotics* 40: 452–471.

Tuna T, Nubert J, Pfreundschuh P, et al. (2024b) Informed, constrained, aligned: a field analysis on degeneracy-aware point cloud registration in the wild. ArXiv Preprint arXiv: 2408.11809.

Wang Y, Song W, Zhang Y, et al. (2024) Four years of multimodal odometry and mapping on the rail vehicles. *Journal of Field Robotics* 41(2): 227–257.

Wei H, Jiao J, Hu X, et al. (2024) Fusionportablev2: a unified multi-sensor dataset for generalized slam across diverse platforms and scalable environments. ArXiv Preprint arXiv:2404.08563.

Xu W, Cai Y, He D, et al. (2021) Fast-lio2: fast direct lidar-inertial odometry. *IEEE Transactions on Robotics* 38: 2053–2073.

Zhang J, Kaess M and Singh S (2016) On degeneracy of optimization-based state estimation problems. In: 2016 IEEE international conference on robotics and automation (ICRA), Stockholm, Sweden, 16–21 May 2016, 809–816.

Zhang L, Helmberger M, Fu LFT, et al. (2022) Hilti-Oxford dataset: a millimeter-accurate benchmark for simultaneous localization and mapping. *IEEE Robotics and Automation Letters* 8(1): 408–415.

Zhang Y, Tian Y, Wang W, et al. (2023) Ri-lio: reflectivity image assisted tightly-coupled lidar-inertial odometry. *IEEE Robotics and Automation Letters* 8: 1802–1809.

Zhao S, Gao Y, Wu T, et al. (2024) Subt-mrs dataset: pushing slam towards all-weather environments. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR), Seattle, WA, 16–22 June 2024, pp. 22647–22657.

Zheng C, Zhu Q, Xu W, et al. (2022) Fast-livo: fast and tightly-coupled sparse-direct lidar-inertial-visual odometry. In: 2022 IEEE/RSJ international conference on intelligent robots and systems (IROS), Kyoto, Japan, 23–27 October 2022, 4003–4009. IEEE.

Zhu F, Ren Y and Zhang F (2022) Robust real-time lidar-inertial initialization. In: 2022 IEEE/RSJ international conference on intelligent robots and systems (IROS), Kyoto, Japan, 23–27 October 2022, 3948–3955.